Table 1: Sequential MAS results for Qwen3-8B. Acc. is accuracy (%), Token and Time values for Single, TextMAS, and Linear are taken directly from the reference figures, where Time is labeled Speed (A100). The remaining methods retain their H200 measurements. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by −.

Tasks	**Metrics**	**Baselines**		**LatentMAS variants**					**Improve**
		Single	TextMAS	Identical	Linear	**Kernel**	**Kernel-ES**	Soft	
Sequential MAS Setting									
AIME24	Acc.	50.0	53.3	–	56.7	**66.25**	64.17	63.75	↑ **13.0**
	Token	12891	38596	–	8953	**8548**	8770	22342	↓ **77.9%**
	Time	421	2808	–	688	**225.6**	225.8	549.0	× **4.5**
AIME25	Acc.	46.7	**53.3**	–	**53.3**	49.58	50.42	49.17	↓ 2.9
	Token	14692	45088	–	**8699**	10158	9830	24557	↓ **78.2%**
	Time	450	3150	–	**820**	261.2	255.5	597.0	× **4.5**
ARC-C	Acc.	91.0	94.6	94.54	94.4	94.80	**95.56**	–	↑ **1.0**
	Token	846	2252	559	**529**	570	915	–	↓ **74.7%**
	Time	266	2059	3.7	**703**	3.7	4.5	–	× **3.9**
ARC-E	Acc.	95.6	**99.1**	98.27	98.8	98.32	98.40	–	↓ 0.7
	Token	656	2085	**468**	490	472	822	–	↓ **77.4%**
	Time	404	3702	**2.5**	1759	**2.5**	3.2	–	× **4.4**
GPQA	Acc.	39.9	43.4	–	45.5	**48.74**	48.61	–	↑ **5.3**
	Token	6435	17986	–	4571	**3588**	3846	–	↓ **80.1%**
	Time	813	5771	–	854	**58.3**	59.4	–	× **5.0**
GSM8K	Acc.	81.1	92.3	91.58	**93.8**	91.74	92.19	–	↓ 0.1
	Token	1280	2324	**628**	860	642	1029	–	↓ **72.4%**
	Time	449	1739	**4.1**	543	**4.1**	5.2	–	× **3.7**
HumanEval+	Acc.	74.4	80.5	**84.15**	80.5	83.54	83.38	82.62	↑ **3.0**
	Token	2507	4593	1668	1866	**1637**	1696	15055	↓ **64.4%**
	Time	502	1619	18.9	497	**18.8**	21.5	112.4	× **2.8**
MBPP+	Acc.	64.8	69.5	–	74.6	**75.26**	75.20	73.15	↑ **5.8**
	Token	2053	3695	–	**1164**	1462	1528	14388	↓ **60.4%**
	Time	1064	3628	–	**1275**	17.9	19.6	101.6	× **2.5**
MedQA	Acc.	53.0	75.0	**76.33**	75.3	73.67	72.00	66.67	↓ 1.3
	Token	2098	4260	1872	1555	**1139**	1154	15972	↓ **73.3%**
	Time	476	1923	17.0	928	**12.5**	13.6	111.1	× **3.7**

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 2: Hierarchical MAS results for Qwen3-8B. Acc. is accuracy (%), Token and Time values for Single, TextMAS, and Linear are taken directly from the reference figures, where Time is labeled Speed (A100). The remaining methods retain their H200 measurements. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by −.

Tasks	**Metrics**	**Baselines**		**LatentMAS variants**					**Improve**
		Single	TextMAS	Identical	Linear	**Kernel**	**Kernel-ES**	Soft	
Hierarchical MAS Setting									
AIME24	Acc.	50.0	53.3	–	53.3	60.00	**65.83**	–	↑ **12.5**
	Token	12891	42629	–	**7526**	9926	9898	–	↓ **76.8%**
	Time	421	3132	–	**776**	258.4	257.9	–	× **4.3**
AIME25	Acc.	46.7	50.0	–	50.0	**53.33**	52.50	–	↑ **3.3**
	Token	14692	53929	–	13230	**11207**	11300	–	↓ **79.2%**
	Time	450	3488	–	616	**289.1**	291.9	–	× **4.8**
ARC-C	Acc.	91.0	93.3	94.20	93.9	**95.05**	**95.05**	–	↑ **1.8**
	Token	846	2854	606	**344**	603	822	–	↓ **78.9%**
	Time	266	2034	3.9	**714**	3.8	4.3	–	× **4.7**
ARC-E	Acc.	95.6	98.2	**98.48**	98.3	98.23	98.23	–	↑ **0.0**
	Token	656	2237	494	**308**	495	721	–	↓ **77.9%**
	Time	404	3619	2.6	**1779**	2.5	3.0	–	× **4.6**
GPQA	Acc.	39.9	43.0	–	46.9	**51.39**	49.75	–	↑ **8.4**
	Token	6435	22450	–	**3395**	3752	3964	–	↓ **83.3%**
	Time	813	6108	–	**798**	59.9	61.5	–	× **6.0**
GSM8K	Acc.	81.1	90.4	91.58	89.5	91.28	**92.34**	–	↑ **1.9**
	Token	1280	2370	688	**353**	704	932	–	↓ **70.3%**
	Time	449	1365	4.1	**702**	4.4	4.8	–	× **3.3**
HumanEval+	Acc.	74.4	76.8	82.62	78.0	**83.84**	83.23	–	↑ **7.0**
	Token	2507	8768	1688	**1274**	1681	1637	–	↓ **81.3%**
	Time	502	1809	20.2	**439**	20.1	21.0	–	× **5.2**
MBPP+	Acc.	64.8	71.9	70.97	72.2	72.75	72.16	**73.21**	↑ **0.8**
	Token	2053	7703	1436	**1264**	1502	1460	12768	↓ **81.0%**
	Time	1064	3898	20.3	**1387**	20.1	20.2	92.2	× **5.1**
MedQA	Acc.	53.0	76.3	75.33	**77.0**	73.33	72.67	–	↓ 3.0
	Token	2098	6893	1391	**1007**	1351	1336	–	↓ **80.6%**
	Time	476	3387	15.5	**964**	15.7	14.8	–	× **5.4**

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 3: Sequential MAS results for Qwen3-14B. Acc. is accuracy (%), Token and Time values for Single, TextMAS, and Linear are taken directly from the reference figures, where Time is labeled Speed (A100). The remaining methods retain their H200 measurements. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by −.

Tasks	**Metrics**	**Baselines**		**LatentMAS variants**					**Improve**
		Single	TextMAS	Identical	Linear	**Kernel**	**Kernel-ES**	Soft	
Sequential MAS Setting									
AIME24	Acc.	63.3	63.3	–	66.7	**72.08**	69.58	–	↑ **8.8**
	Token	11263	32092	–	10593	**8259**	8489	–	↓ **74.3%**
	Time	1018	4554	–	1149	**255.6**	283.4	–	× **3.9**
AIME25	Acc.	56.7	60.0	–	**63.3**	56.67	60.00	49.58	↑ **0.0**
	Token	11298	44618	–	11402	**10266**	10402	21403	↓ **77.0%**
	Time	1040	5184	–	1473	**335.5**	350.6	575.5	× **4.3**
ARC-C	Acc.	92.6	95.9	**95.99**	95.6	95.48	95.73	–	↓ 0.2
	Token	773	2985	463	**426**	461	544	–	↓ **84.6%**
	Time	338	5125	4.3	**1136**	4.3	6.0	–	× **6.5**
ARC-E	Acc.	97.2	99.0	98.48	**99.4**	98.82	98.57	–	↓ 0.2
	Token	608	1670	379	**224**	380	462	–	↓ **77.3%**
	Time	551	9171	2.8	**2124**	2.7	4.5	–	× **4.4**
GPQA	Acc.	48.5	51.5	–	52.0	**55.81**	54.42	53.54	↑ **4.3**
	Token	5547	12676	–	5454	3009	**2995**	18528	↓ **76.4%**
	Time	1043	9714	–	1475	**59.4**	67.2	222.9	× **4.2**
GSM8K	Acc.	83.7	93.8	92.12	**95.2**	93.10	93.33	–	↓ 0.5
	Token	1118	3324	577	644	**571**	643	–	↓ **82.8%**
	Time	536	3729	**5.0**	1952	**4.7**	6.7	–	× **5.8**
HumanEval+	Acc.	76.8	81.1	86.89	86.5	88.11	86.43	**90.70**	↑ **7.0**
	Token	2366	5934	**1590**	2042	1594	1607	19587	↓ **73.1%**
	Time	1084	4062	**24.1**	1285	**24.1**	28.1	157.3	× **3.7**
MBPP+	Acc.	68.5	72.8	78.17	75.7	78.11	**79.43**	–	↑ **6.6**
	Token	1858	4971	1345	1621	**1336**	1346	–	↓ **73.1%**
	Time	2410	8728	21.9	2400	**21.5**	25.5	–	× **3.7**
MedQA	Acc.	64.7	80.3	81.00	80.7	**81.67**	79.33	79.00	↑ **1.4**
	Token	1746	3444	1616	1841	922	**912**	22875	↓ **73.5%**
	Time	1360	4142	19.0	1420	**13.2**	14.3	151.0	× **3.8**

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.

Table 4: Hierarchical MAS results for Qwen3-14B. Acc. is accuracy (%), Token and Time values for Single, TextMAS, and Linear are taken directly from the reference figures, where Time is labeled Speed (A100). The remaining methods retain their H200 measurements. Best results in each row are bold. Kernel and Kernel-ES are highlighted. Improve compares the better of Kernel and Kernel-ES with TextMAS. Missing experiment combinations are denoted by −.

Tasks	Metrics	Baselines		LatentMAS variants					Improve
		Single	TextMAS	Identical	Linear	**Kernel**	**Kernel-ES**	Soft	
Hierarchical MAS Setting									
AIME24	Acc.	63.3	70.0	–	**73.3**	69.17	69.58	71.25	↓ 0.4
	Token	11263	29025	–	10230	**8947**	9017	23924	↓ **69.2%**
	Time	1018	5718	–	1089	**284.1**	291.0	639.9	× **3.2**
AIME25	Acc.	56.7	**66.7**	–	**66.7**	59.17	57.92	–	↓ 7.5
	Token	11298	50003	–	**9527**	10755	11122	–	↓ **78.5%**
	Time	1040	6019	–	**1056**	346.9	365.7	–	× **4.6**
ARC-C	Acc.	92.6	95.3	95.65	95.5	**95.99**	95.65	–	↑ **0.7**
	Token	773	2167	434	**295**	424	477	–	↓ **80.4%**
	Time	338	4283	4.0	**1090**	3.7	5.4	–	× **5.1**
ARC-E	Acc.	97.2	98.3	98.70	98.7	**98.78**	98.23	–	↑ **0.5**
	Token	608	2752	344	619	**337**	402	–	↓ **87.8%**
	Time	551	7102	2.5	1884	**2.4**	4.0	–	× **8.2**
GPQA	Acc.	48.5	52.0	–	53.0	54.80	**55.81**	54.55	↑ **3.8**
	Token	5547	20931	–	3606	2769	**2755**	20909	↓ **86.8%**
	Time	1043	9119	–	1458	**61.5**	63.6	287.8	× **7.6**
GSM8K	Acc.	83.7	90.8	91.96	91.6	91.81	**92.49**	–	↑ **1.7**
	Token	1118	3021	574	495	**569**	602	–	↓ **81.2%**
	Time	536	3675	4.6	**1631**	**4.6**	6.0	–	× **5.3**
HumanEval+	Acc.	76.8	84.1	88.41	86.6	87.04	86.74	**91.16**	↑ **2.9**
	Token	2366	8114	1473	1512	1454	**1430**	21403	↓ **82.4%**
	Time	1084	3988	**23.8**	1188	**24.0**	25.4	185.6	× **5.7**
MBPP+	Acc.	68.5	73.0	76.92	73.8	78.04	**78.31**	–	↑ **5.3**
	Token	1858	7458	1209	1187	**1132**	1182	–	↓ **84.8%**
	Time	2410	9162	23.3	**2507**	**21.6**	23.7	–	× **6.6**
MedQA	Acc.	64.7	78.0	**81.33**	78.3	79.00	77.67	48.33	↑ **1.0**
	Token	1746	5473	870	899	**860**	875	23700	↓ **84.3%**
	Time	1360	7591	**12.1**	1250	**12.4**	13.8	173.7	× **6.4**

Note. MedQA may be removed. Repetition rule: datasets with 1–100 problems are repeated 8 times; those with 101–1000 problems, 4 times; and those with more than 1000 problems, once.